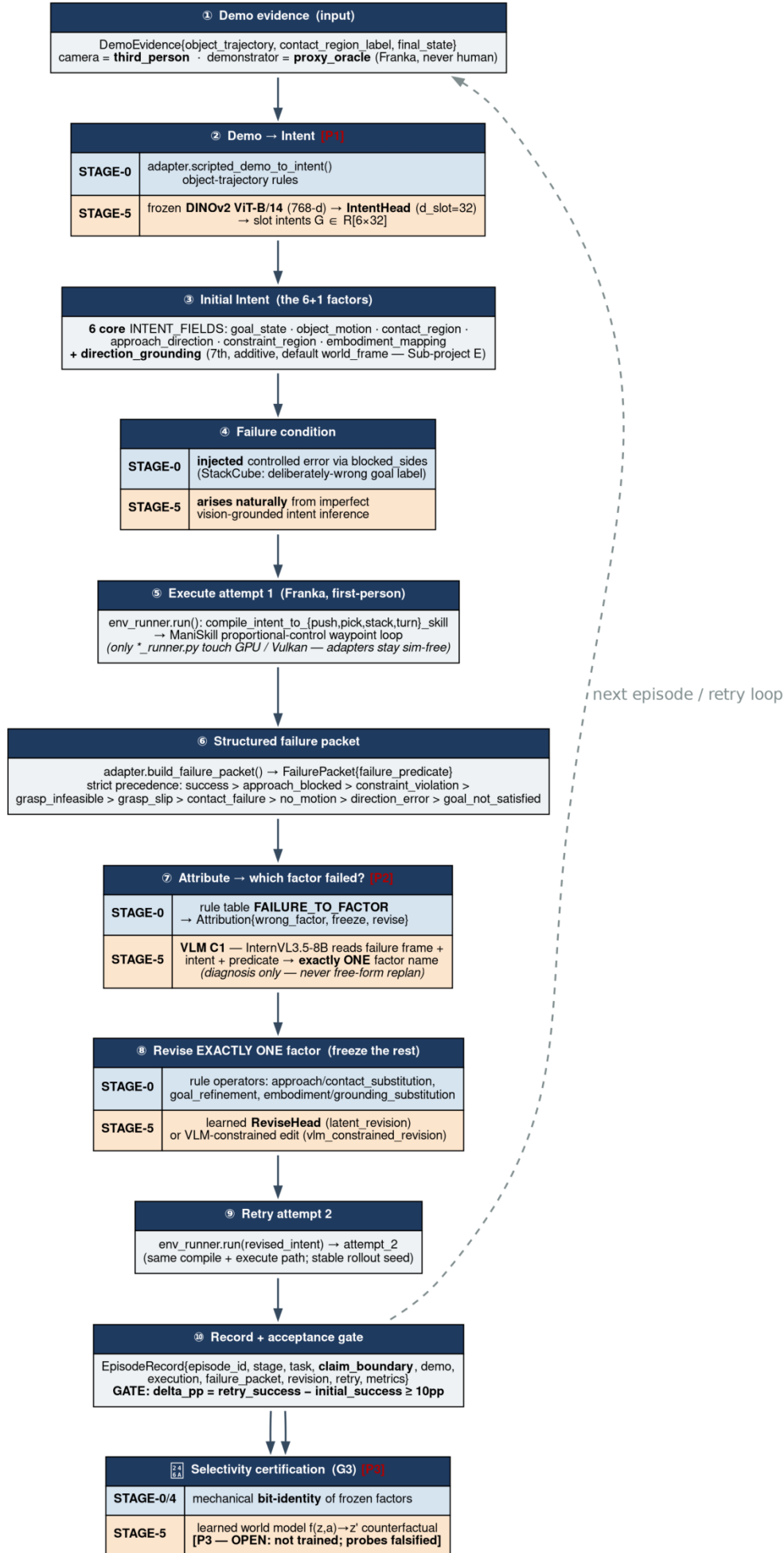


BABYSTEPS — failure-guided single-factor intent-revision loop

blue = Stage-0 discrete/scripted path | orange = Stage-5 vision-grounded / VLM overlay | P4 = learned action decoder (replaces skill compiler, DEFERRED)



Data contract & working invariants

Source of truth = *babysteps/schemas.py* whitelists; *goal.md* authoritative over docs where they disagree.

The six core + one additive intent factors (object-centric, task-agnostic)

factor	meaning	example tokens (schemas.py whitelist)
goal_state	desired final object relation/pose	cube_at_target · cubeA_on_cubeB · faucet_turned
object_motion	observed/intended object movement	translate_+x · lift_up · place_on · turn
contact_region	demonstrated/inferred contact site	minus_x_face · handle_grip · faucet_base
approach_direction	route/side used to reach contact	from_minus_x · from_above
constraint_region	scene/state to preserve	none · faucet_base_static
embodiment_mapping	proxy contact -> Franka action	..._push · ..._grasp · ..._poke_turn
direction_grounding	(7th, additive) demo-direction frame	actor_frame · observer_frame · world_frame

Confirmed invariants (verified against code)

- **Single-factor revision.**
every revision changes exactly one INTENT_FIELDS factor; Revision.frozen_factors lists the rest; enforced via dataclasses.replace(). (One legacy 2-factor exception: constraint_introduction— see verification page.)
- **Additive schema.**
add new tokens + stop emitting old ones; defer removal to a cleanup pass. direction_grounding is the 7th field, outside INTENT_FIELDS, serialized only when non-default.
- **Demo = object evidence.**
demo describes object motion, never an executable Franka motor program; camera=third_person, demonstrator=proxy_oracle.
- **Sim privilege boundary.**
only *_runner.py import ManiSkill/Vulkan; the demo->intent path and tests/ stay sim-free (login-node runnable).
- **One orchestrator.**
Stage-0 and Stage-5 share run_episode() and the same EpisodeRecord JSON; claim_boundary stamped on every record.
- **Universal gate.**
 $\text{delta_pp} = \text{retry_success} - \text{initial_success} \geq 10\text{pp}$, across Stage-0 discrete and Stage-4/5 latent revision alike.
- **VLM = diagnosis only.**
the VLM picks one factor name (which?), never free-form replanning (what value?). C2 free-form is a baseline only.

Sub-projects (A-E) & Stage-5 status

All five pass the $\Delta_{pp} \geq 10$ acceptance gate in Stage-0.

Five Stage-0 task families — each revises a different single factor

ID	task	revised factor	inference failure mode
A	PushCube-v1	contact_region / approach_dire	demo viewpoint conflates which face was contacted
B	PickCube-v1	contact_region	grasp face occluded in demo; grasp_slip at runtime
C	StackCube-v1	goal_state	demo goal-ambiguous (place-near vs stack-on); failure in label, no
D	TurnFaucet-v1	embodiment_mapping	demo grasp-turn infeasible on Franka wrist range (poke_turn unrel
E	CrossViewPush-v1	direction_grounding	demo camera != robot camera -> direction-frame mismatch (7-fa

Stage-5 priorities (status as of 2026-06-01)

- **P1 vision swap — PARTIAL.**
PushCube end-to-end PASSES: held-out seeds 100-149 latent 48/50 (0.96) vs oracle 49/50 (0.98); G4 +96pp, G5 -2pp, object_motion probe 0.95. StackCube G1 FAILS: relational object_motion 0.42 (<0.90) across all pooling — scope narrowed to PushCube-only.
- **P2 VLM attribution — DONE.**
5-task main table passes all 3 gates (C1 attr $\geq$ rule-table; C1 pres $\geq$ C2; C1 succ within 5pp of C2). C1 attr: Push 1.0 · Pick 1.0 · Stack 0.86 · Turn 1.0 · CrossView 1.0; +92pp success on PickCube. StackCube 0.86 = residual open issue (7 misses on object_motion).
- **P3 world model — OPEN.**
no learned world model trained; G3 selectivity still mechanical bit-identity. Only de-risking probes run (VLM-reads-from-demo, top-down vision) — ALL falsified.
- **P4 action decoder — DEFERRED.**
optional; paper stands on P1-P3 + skill compilers.
- **M2b pixel encoder — OPEN.**
not a named milestone; most likely the P1 vision_features.py pixel path, itself blocked on the StackCube relational bottleneck.
- **Done supporting:**
M1 locked claim · M2 5-task Stage-0 · M3 7 procedural baselines (Push 0.98/Pick 0.92/Stack 0.70) · M6 related-work. M4 main table + M7 paper PENDING P3.

Misalignments to double-check

Verified against *schemas.py* / *revision.py* / *failure.py* / *episode.py* / *vlm_attribute.py* on 2026-06-01.

Items resolved 2026-06-01 are listed first; then still-open repo items, then extractor noise where the CODE IS ALREADY CORRECT (listed so you don't chase false alarms).

Resolved 2026-06-01 (docs updated this session)

- **VLM model identity.**
docs now state InternVL3.5-8B (method VLM-agnostic) across goal.md, CLAUDE.md, milestones.md, update.md and the locked-claim doc.
- **goal.md P2 status refreshed.**
stale v1 (3 tasks, StackCube 0%) replaced with the 5-task main table — all gates pass, StackCube attr 0.86, PickCube +92pp.
- **'six' -> '6 core + 1 additive'.**
CLAUDE.md prose now matches its own 7-row table (direction_grounding = additive 7th, Sub-project E).
- **goal.md example tokens.**
shorthand translate_right / left_face / from_left replaced with real schema tokens translate_+x / minus_x_face / from_minus_x.
- **latent-intent framing.**
new 'Framing: latent intent vs. the swappable diagnoser' subsection added to goal.md Stage-5 (addresses the headline-scope risk below).

A. Doc <-> code drift (remaining)

- **[LOW] claim_boundary phrasing.**
docs paraphrase it; the actual token is CLAIM_BOUNDARY='third_person_demo_proxy_not_human_demo' (schemas.py). Cosmetic — left as-is.

B. Claim / scope risks (paper-facing)

- **[HIGH] '5-task vision-grounded' headline.**
P1 latent G is end-to-end ONLY on PushCube (StackCube probe 0.42, scope narrowed). P2 VLM attribution runs on failure FRAMES + predicate strings, not on latent G — so 4/5 tasks use the Stage-4 discrete scaffold. Now framed honestly in goal.md ('latent' = representation + slot-local revision, demonstrated end-to-end on PushCube); hold this line in the paper draft.
- **[MED] single-factor invariant vs constraint introduction.**
constraint_introduction (TurnFaucet, now deprecated in favour of embodiment_substitution) revises constraint_region AND contact_region — a 2-factor operator whose docstring still says single-factor. frozen_factors accounts for it, but legacy_compute_metrics could mis-handle len==2. The headline invariant has one live exception.

C. Code gaps / cruft

- **[MED] continuous_rotation_infeasible has no rule.**
predicate is in FAILURE_PREDICATES (schemas.py) but absent from FAILURE_TO_FACTOR (failure.py) — attribute_failure() would raise ValueError if it is ever emitted (TurnFaucet redesign in flight).
- **[LOW] silent learned-policy degradation.**
episode.py comment says provider=None makes latent_revision 'degrade to a no-op'; in fact the default policy just ignores demo_features with no warning/metric. Add an explicit guard/log.
- **[LOW] deprecated tokens linger.**
proxy_contact_to_franka_turn, constraint_violation, constraint_introduction kept in whitelists per the additive rule; schedule the cleanup pass once tests prove no references.

D. Extractor artifacts — CODE IS FINE (ignore)

- **Revision field names.**
an extractor listed Revision={wrong_factor, operator_name}; the real dataclass is {operator, factor, old_value, new_value, frozen_factors} — code is correct, wrong_factor lives on Attribution.
- **Operator names.**
an extractor wrote goal_substitution / constraint_substitution; the real operators are goal_refinement / constraint_introduction — code is correct.
- **Enum membership.**
an extractor gave 3 direction_groundings & omitted faucet_base; code has 4 groundings (+object_frame, reserved) and 6 contact_regions — code is correct.